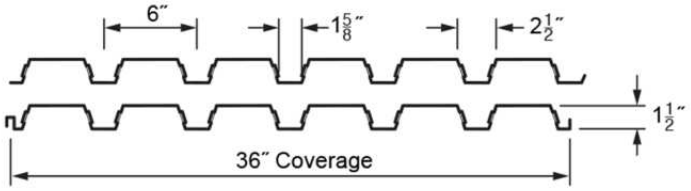
1.5CD, 1.5CDI Composite Deck

NORMAL-WEIGHT CONCRETE

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	36	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



ICC-ES APPROVED
REFER TO ESR-3818

ASD SECTION PROPERTIES

DESIGN STRENGTHS (No Concrete Fill)

Gage	F_y (ksi)	A_s (in^2/ft)	I_D 1 Span (in^4/ft)	I_D 2+ Span (in^4/ft)	I_p (in^4/ft)	I_n (in^4/ft)	S_p (in^3/ft)	S_n (in^3/ft)	Mn,p/Ω (in-lb/ft)	Mn,n/Ω (in-lb/ft)	Vn/Ω (lb/ft)	*Rbe/Ω (lb/ft)	*Rbi/Ω (lb/ft)
22	40	0.480	0.163	0.175	0.157	0.175	0.176	0.183	4215	4391	2009	654	1181
20	40	0.582	0.205	0.213	0.201	0.213	0.224	0.235	5375	5624	2428	932	1696
18	40	0.770	0.280	0.281	0.279	0.281	0.304	0.315	7291	7553	3192	1557	2858
16	40	0.971	0.355	0.355	0.355	0.355	0.390	0.395	9332	9454	3996	2381	4396

CONSTRUCTION CLEAR SPANS

Total Slab Depth (in)	Gage	Concrete Weight (psf)	Maximum Construction Clear Span (ft - in)				Concrete Volume (ft^3/ft^2)
			Single	Double	Triple	Cantilever	
3 1/2	22	31	5 - 9	6 - 9	6 - 10	2 - 0	0.213
	20		6 - 9	8 - 0	8 - 1	2 - 5	
	18		8 - 4	9 - 8	9 - 11	3 - 1	
	16		9 - 8	10 - 9	11 - 1	3 - 8	
4	22	37	5 - 5	6 - 5	6 - 6	1 - 11	0.255
	20		6 - 5	7 - 7	7 - 8	2 - 4	
	18		7 - 11	9 - 2	9 - 4	3 - 0	
	16		9 - 1	10 - 2	10 - 6	3 - 6	
4 1/2	22	43	5 - 3	6 - 2	6 - 3	1 - 10	0.297
	20		6 - 2	7 - 3	7 - 4	2 - 3	
	18		7 - 6	8 - 9	8 - 10	2 - 10	
	16		8 - 8	9 - 8	10 - 1	3 - 4	
5	22	49	5 - 0	5 - 11	5 - 11	1 - 10	0.338
	20		5 - 11	6 - 11	7 - 0	2 - 3	
	18		7 - 2	8 - 4	8 - 6	2 - 9	
	16		8 - 4	9 - 3	9 - 7	3 - 2	
5 1/2	22	55	4 - 10	5 - 8	5 - 9	1 - 9	0.380
	20		5 - 8	6 - 8	6 - 9	2 - 2	
	18		6 - 11	8 - 0	8 - 2	2 - 8	
	16		8 - 0	8 - 11	9 - 3	3 - 1	
6	22	61	4 - 8	5 - 6	5 - 7	1 - 9	0.422
	20		5 - 6	6 - 5	6 - 6	2 - 1	
	18		6 - 8	7 - 9	7 - 10	2 - 7	
	16		7 - 9	8 - 7	8 - 11	2 - 11	

Normal-Weight Concrete (145 pcf)

Notes:

1. Section properties are calculated in accordance with the AISI S100-16.
2. Web crippling design strengths* and maximum construction spans are based on 1 1/2" for end bearing and 3" for interior bearing. Check web crippling if minimums are not met.
3. Maximum construction spans are based on ANSI/SDI C-2017 Standard for Composite Steel Floor Deck and the following construction loading:
   - Deck self-weight and concrete weight plus worst-case of either a 150 lb.concentrated load or a 20 psf uniform load; or
   - Deck self-weight plus a 50 psf uniform construction load, whichever controls.
4. Concrete weights do not include weight of deck.
5. Deck profile has been accounted for in determining concrete volumes. Deck and support deflections have not been included in concrete volumes or weights.
6. Cantilever clear spans assume adjacent clear span is at least 3 times longer than the cantilever clear span and no greater than spans shown. Bearing width at perimeter support assumed to be 3"minimum. Contact New Millennium otherwise.

SLAB DESIGN - ALLOWABLE SUPERIMPOSED UNIFORM LOADS AND MINIMUM SDI SLAB REINFORCEMENT

Normal-Weight Concrete (yc = 145 pcf, f'c = 3000 psi, n = 9)	Total Slab Depth (in)	Gage	Allowable Superimposed Uniform Load (psf) Clear Span (ft - in)									Recommended Welded Wire Fabric	Wire Area (in^2/ft)
			6 - 0	6 - 6	7 - 0	7 - 6	8 - 0	8 - 6	9 - 0	10 - 0	11 - 0		
	3 1/2	22	294	246	207	176	151	130	113	85	65	6x6 - W1.4xW1.4	0.028
		20	354	297	251	215	185	160	139	106	82		
		18	400	386	328	281	243	212	185	139	105		
		16	400	400	400	349	301	251	212	154	116		
	4	22	370	310	262	223	191	165	143	109	83	6x6 - W1.4xW1.4	0.028
		20	400	375	318	272	234	203	177	136	105		
		18	400	400	400	357	309	269	236	183	145		
		16	400	400	400	400	385	336	295	229	172		
	4 1/2	22	400	377	319	272	234	202	175	133	103	6x6 - W1.4xW1.4	0.028
		20	400	400	387	331	286	248	216	167	130		
		18	400	400	400	400	378	330	289	225	178		
		16	400	400	400	400	400	400	363	285	228		
	5	22	400	400	378	322	277	240	208	159	123	6x6 - W2.0xW2.0	0.040
		20	400	400	400	393	339	295	258	199	156		
		18	400	400	400	400	400	392	344	269	213		
		16	400	400	400	400	400	400	400	341	273		
	5 1/2	22	400	400	400	374	322	279	242	186	144	6x6 - W2.0xW2.0	0.040
		20	400	400	400	400	394	343	300	232	182		
		18	400	400	400	400	400	400	400	314	249		
		16	400	400	400	400	400	400	400	398	319		
	6	22	400	400	400	400	367	318	277	212	165	4x4 - W1.4xW1.4	0.042
		20	400	400	400	400	400	392	342	265	208		
		18	400	400	400	400	400	400	400	360	286		
		16	400	400	400	400	400	400	400	400	366		

Notes:

7. Welded wire fabric shown in the table above is the SDI minimum required for temperature and shrinkage (0.00075 x concrete area above top of deck).
8. Allowable Superimposed Uniform Loads shown in table above are based on the following criteria:
   - Unfactored service level loads, determined using SDI design method per ANSI/SDI C-2017 Standard for Composite Steel Floor Deck-Slabs.
   - Single span conditions without negative bending reinforcing over supports.
   - The presence of shear studs have not been considered, design strength may be increased if shear studs are used.
   - Slab deflection is limited to minimum of Clear Span/360 or 1" under service level superimposed loading.